BENGALURU HOUSE PRICE PREDICTION PROJECT REPORT

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1. Introduction

This project focuses on analyzing and predicting **house prices in Bengaluru** using a real-world dataset.
The primary objective is to identify key factors influencing property prices and evaluate the performance of different **machine learning regression models** for accurate price prediction.

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2. Dataset and Problem Statement

- **Dataset Used:** Bengaluru House Price Data (CSV)
- **Problem Type:** Supervised Regression Task
- **Target Variable:** Price (in lakhs)
- **Original Shape:** 13,320 records, 9 features
- **Final Shape (Post-Cleaning):** 7,251 records, 245 features

The goal is to predict house prices based on features such as **location**, **number of bedrooms (BHK)**, **bathrooms**, and **total area**.

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3. Data Cleaning and Exploratory Analysis

Step 1: Data Loading and Initial Inspection

- Loaded the dataset (`bengaluru_house_prices.csv`) into **df1**.
- Displayed its head and shape for an initial overview.

- **Objective:** Understand structure, feature names, and data types before preprocessing.

Step 2: Irrelevant Column Removal

- Dropped unnecessary columns: *area_type*, *availability*, *society*, and *balcony*.
- Created a cleaner dataset **df2** for focused analysis.
- **Purpose:** Simplify modeling by removing redundant information.

Step 3: Missing Value Handling

- Identified missing values in *location*, *size*, and *bath* columns.
- Removed rows with NaN values from **df2**, resulting in **df3**.
- **Result:** Ensured complete data for feature engineering.

Step 4: Feature Engineering (BHK Extraction)

- Extracted numerical **BHK (Bedrooms, Hall, Kitchen)** values from the *size* column.
- Added a new feature **bhk** in **df4**.
- Example: “2 BHK” → bhk = 2

Step 5: Total Square Feet Cleaning

- Processed *total_sqft* to handle range values (e.g., “1100–1200”) by converting them to averages.
- Ensured all entries were numeric.
- Cleaned data stored in **df5**.

Step 6: Price per Square Foot Calculation

- Created a normalized feature **price_per_sqft** using:
  [
  \text{price_per_sqft} = \frac{\text{price} \times 100000}{\text{total_sqft}}
  ]

- Aids in comparative analysis and outlier detection.

Step 7: Location Dimensionality Reduction

- Cleaned *location* names and grouped those with ≤ 10 occurrences into a single **'other'** category.

- **Purpose:** Reduce high cardinality and maintain model efficiency.

Step 8: Outlier Removal (Per BHK Area)

- Removed unrealistic records where **total_sqft / bhk < 300 sqft/BHK**.

- Resulting dataset: **df6**.

Step 9: Outlier Removal (Price per Sqft)

- For each location, computed mean and standard deviation of *price_per_sqft*.

- Removed entries lying outside one standard deviation.

- Resulting dataset: **df7**.

Step 10: Outlier Removal (BHK vs Price Logic)

- Removed cases where a higher BHK property was priced significantly lower per sqft than a smaller BHK property in the same area.

- Resulting dataset: **df8**.

Step 11: Outlier Removal (Bathrooms)

- Removed properties with excessive bathrooms (**bath > bhk + 2**).

- Cleaned dataset stored as **df9**.

Step 12: Final Feature Selection

- Dropped **price_per_sqft** and **size** columns (no longer needed).

- Resulting dataset: **df10**, containing only relevant features.

Step 13: One-Hot Encoding

- Converted the categorical *location* feature into numerical dummy variables.
- Created **df11** and **df12**, dropping the original *location* and the *'other'* dummy column to avoid multicollinearity.

Step 14: Duplicate Row Removal

- Checked and removed duplicate rows from **df12** to ensure data integrity.

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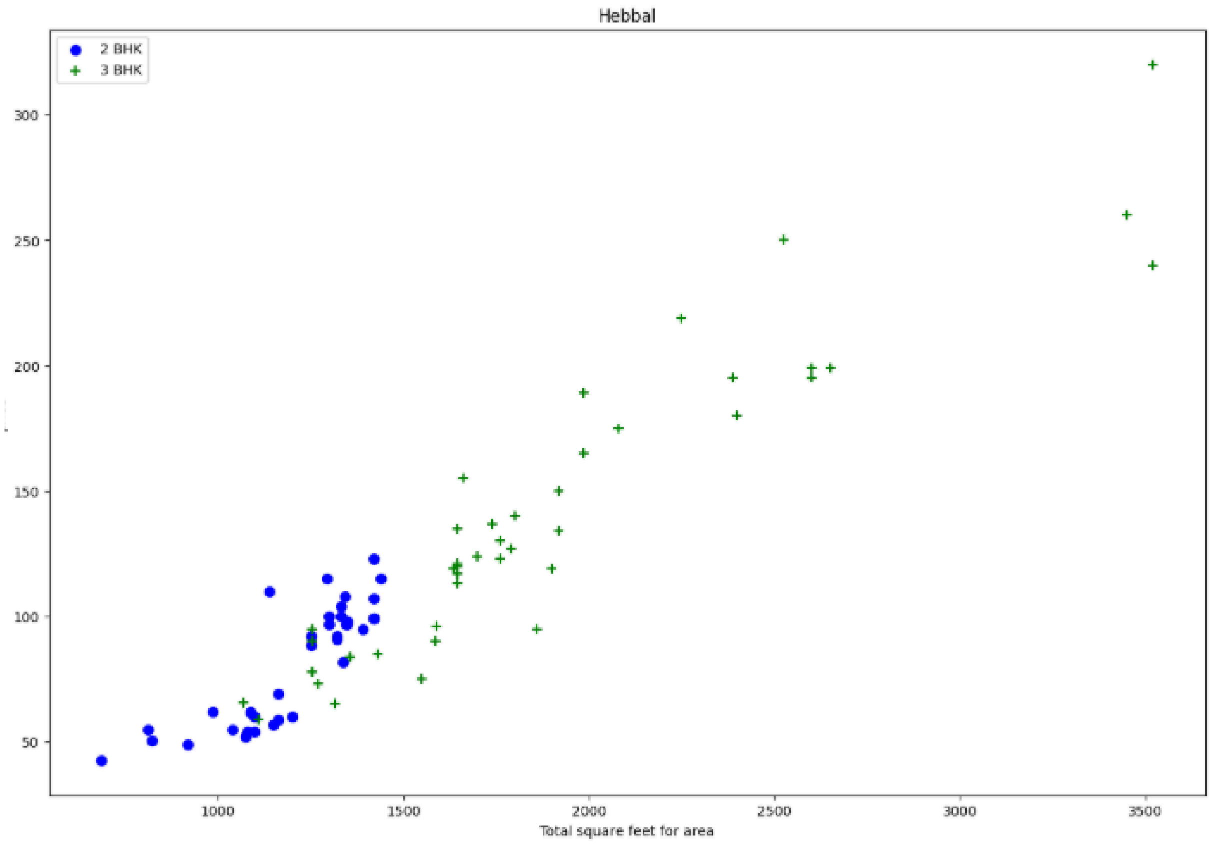
4. Visualization

1. Scatter Plot: 2 BHK vs 3 BHK Price (Before Outlier Removal)

What It Shows: Visualizes 2 BHK (blue) and 3 BHK (green) properties in the *Hebbal* location, plotting *total_sqft* vs *price*.
Key Insight: Reveals inconsistencies where 3 BHK homes appear cheaper than 2 BHK ones — indicating the need for outlier removal.



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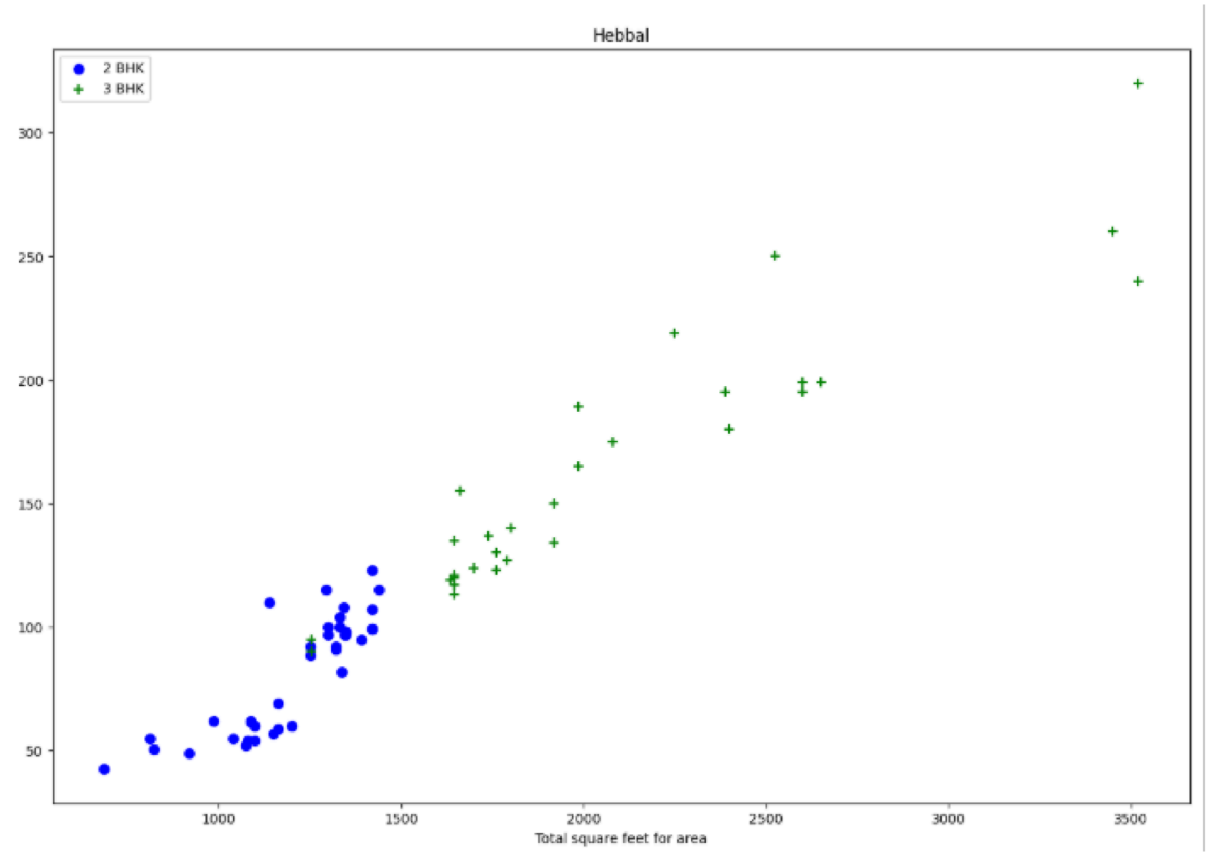
2. Scatter Plot: 2 BHK vs 3 BHK Price (After Outlier Removal)

What It Shows: After applying the cleaning function, the relationship becomes consistent — 3 BHK properties generally cost more than 2 BHK.
Key Insight: Demonstrates that outlier removal improved logical consistency in pricing.



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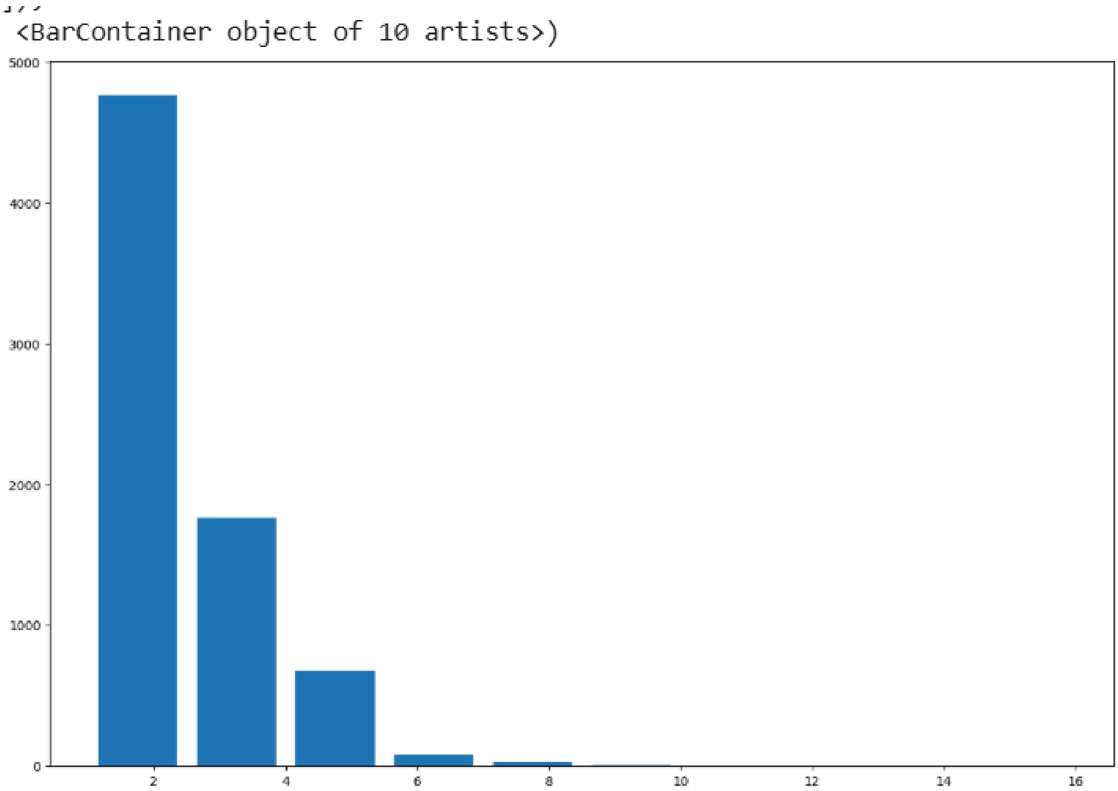
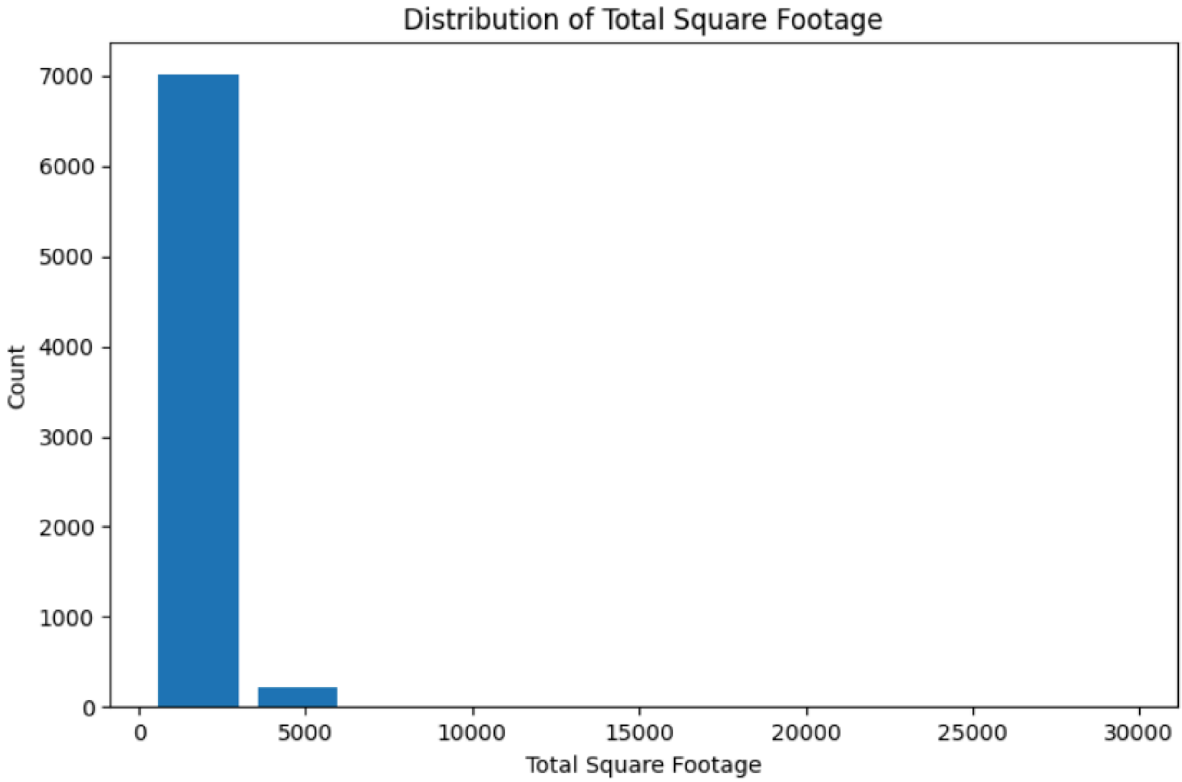
4. Properties with Many Bathrooms

What It Shows: Inspects entries with more than ten bathrooms.
Key Insight: Helped identify unrealistic cases (e.g., 12–16 bathrooms), confirming the need for filtering.

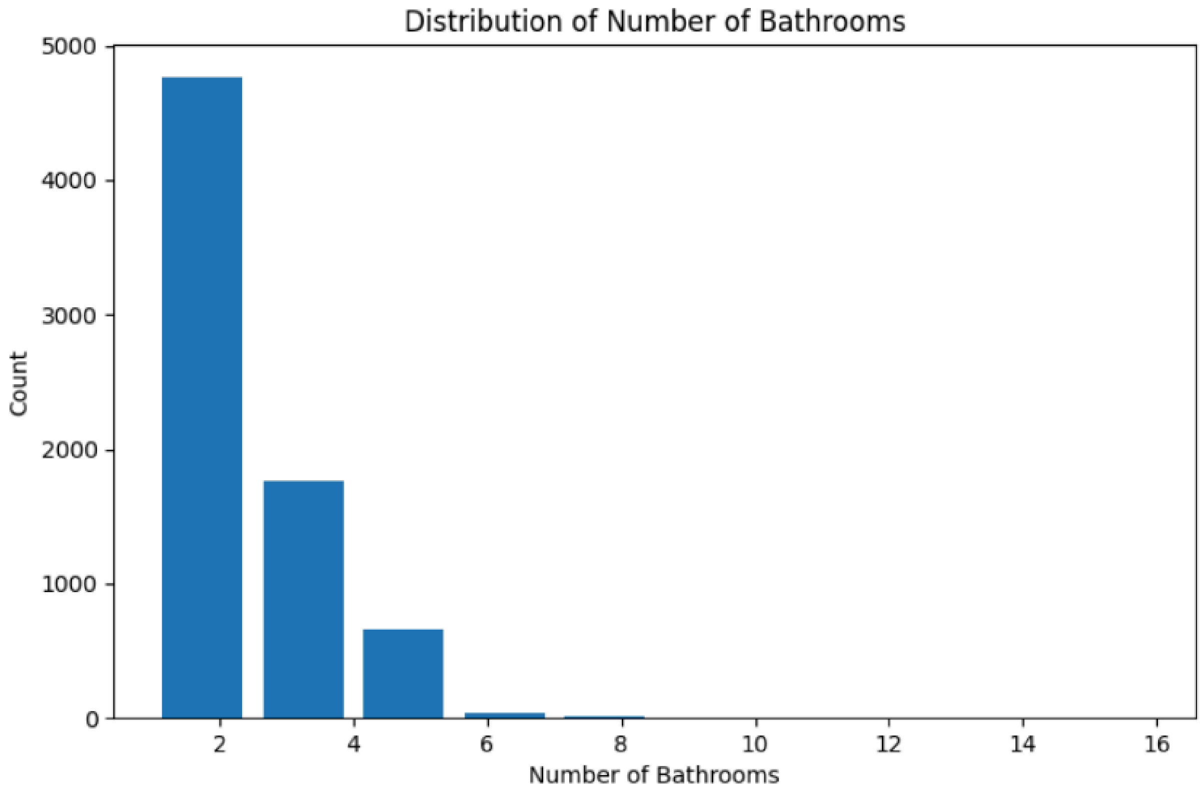


5. Key Insights from Histogram Visualizations

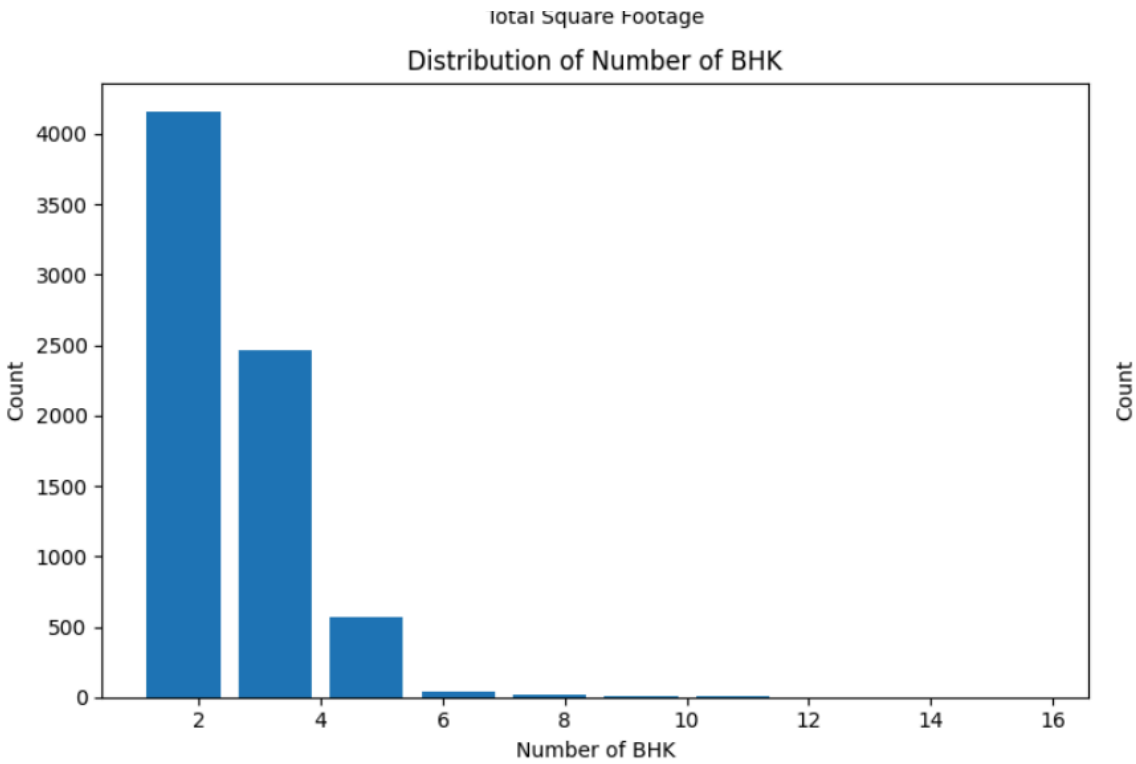
- **Total Square Footage:** Right-skewed; most properties are small, with few large ones.



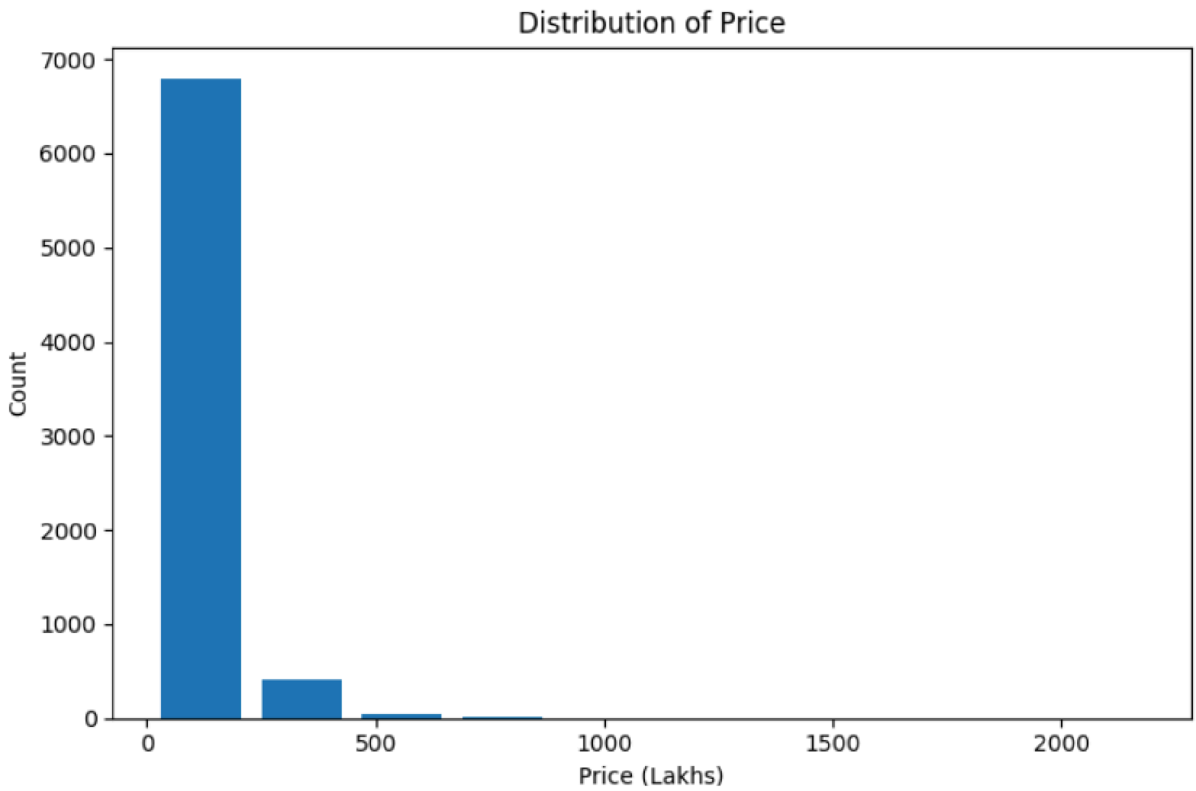
- **Number of Bathrooms:** Most homes have 2 or 3 bathrooms — typical for Bengaluru housing.



- **Number of BHK:** 2 BHK and 3 BHK homes are the most common sizes.



- **Price:** Right-skewed; majority in lower price range, fewer expensive outliers.



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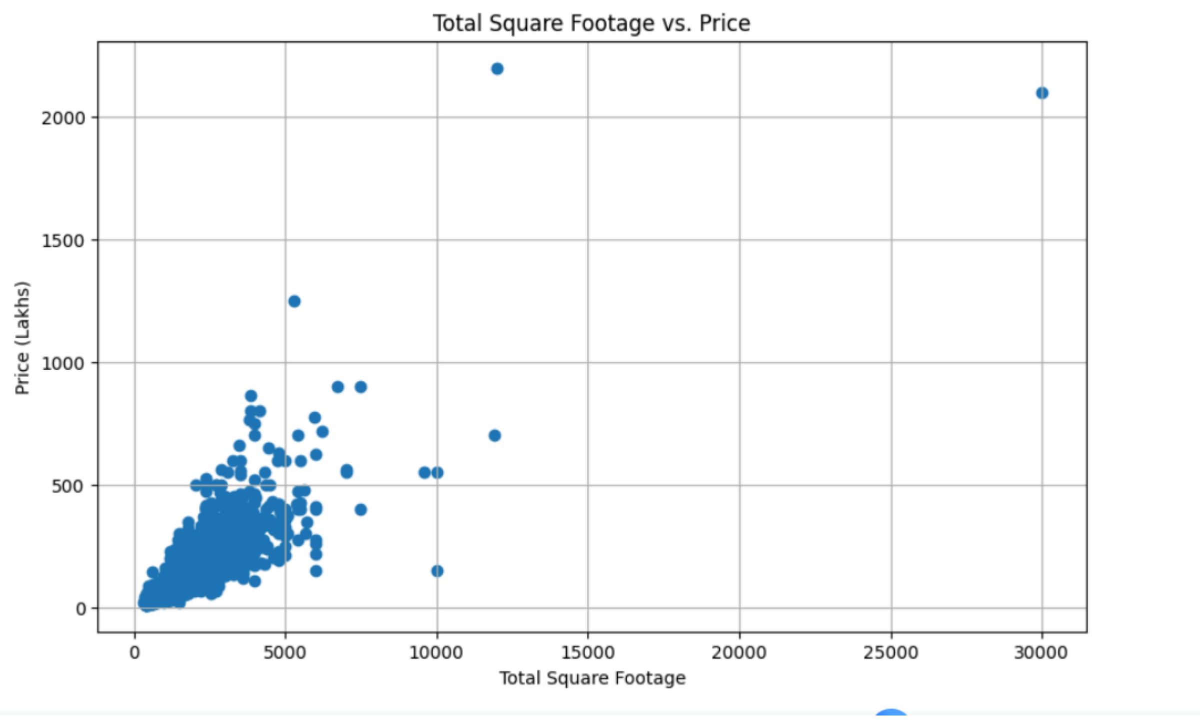
6. Scatter Plot: Total Sqft vs Price (After Cleaning)

What It Shows: Displays a clear positive trend — larger houses generally cost more.
Key Insight: Confirms strong correlation between *total_sqft* and *price*, validating model readiness.



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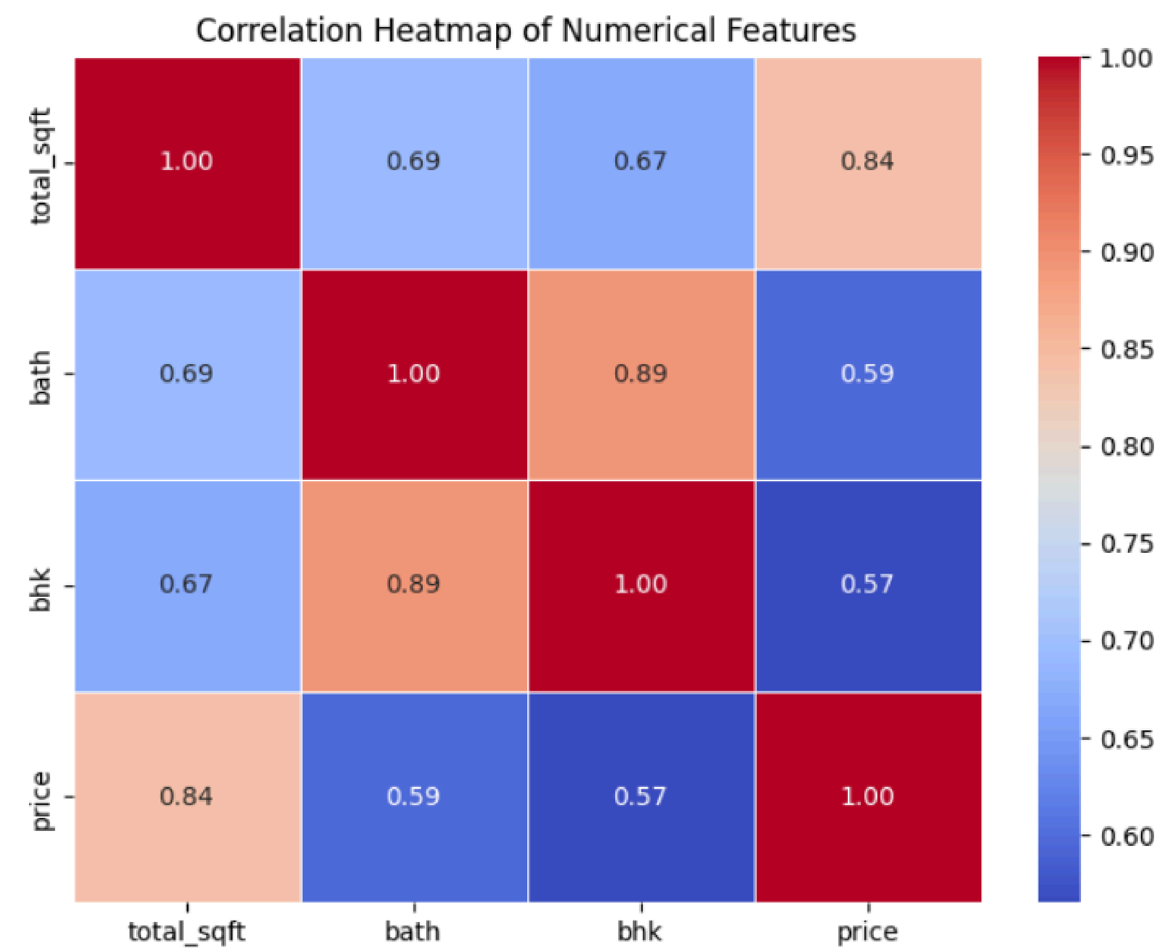
7. Correlation Heatmap of Numerical Features

What It Shows: Pearson correlation among *total_sqft*, *bath*, *bhk*, and *price*.
Key Insight: *total_sqft* shows the highest positive correlation (~0.84) with *price*, followed by *bath* and *bhk*, confirming these as strong predictors.



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5. Preprocessing for Modeling

1. Separating Features and Target Variable

- **X (Features):** All independent variables used for prediction (created by dropping *price*).
- **y (Target):** The *price* column — the value to be predicted.
  Purpose: Ensures the model learns from features without directly seeing the target during training.

2. Train–Test Split

- Dataset divided into **80% training** and **20% testing** subsets.
- **Training set:** Used to train the model.

- **Testing set:** Used to evaluate model performance.
  Purpose: Checks the model's ability to generalize to unseen data and prevents overfitting.

3. Encoding Categorical Features

- Applied **One-Hot Encoding** to the *location* column, converting categorical data into binary columns.

- Created 241 dummy variables representing different locations.
  Purpose: Allows algorithms to understand non-numeric location information without assuming order.

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6. Modeling and Evaluation

1. Modeling (Linear Regression)

A **Linear Regression** model was built to find the best straight-line relationship between house features (*total_sqft*, *bath*, *bhk*) and *price*.
During training, the model learned how each feature affects price.

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2. Evaluation (Model Performance)

a. R^2 Score

- Indicates how much of the price variation is explained by the model's features.

- Achieved an **R^2 score of ≈ 84.5%**, showing strong predictive accuracy.

b. Cross-Validation

- Model tested multiple times with different data splits.

- Consistent scores (**[0.845, 0.800, 0.886]**) show the model is **stable** and **not overfitting**.

c. Model Comparison

Model	R^2 Score	Best Parameters

Linear Regression	0.847 (84.7%)	None
Decision Tree	0.649 (64.9%)	{'criterion': 'friedman_mse', 'splitter': 'best'}

Result: The **Linear Regression** model performed far better, fitting the data more accurately than the Decision Tree Regressor.

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3. Price Prediction (Using the Model)

A **prediction function** was created to estimate house prices based on:

- **Location**
- **Square footage (sqft)**
- **Number of bathrooms (bath)**
- **Number of bedrooms (BHK)**

The inputs are converted into a numerical format compatible with the model (using one-hot encoding), then passed to the trained Linear Regression model to produce a **predicted house price**.
In simple terms, the model uses learned patterns to estimate the most likely price for new properties.

Evaluation Summary

- **Metric Used:** R^2 Score (Coefficient of Determination)
- **Validation Method:** GridSearchCV with 5-fold cross-validation
- **Best Model:** Linear Regression — outperformed Decision Tree by ~20%

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7. Conclusion and Key Insights

This project successfully built a **Bengaluru house price prediction model** through a structured data science process.

Beginning with **data cleaning and exploratory analysis**, the dataset was refined for accuracy and consistency.
Visual insights confirmed strong relationships between **property size**, **bathrooms**, **BHK**, and **price**.

After removing outliers and applying **one-hot encoding**, the data was ready for modeling.
A **comparative evaluation using GridSearchCV** revealed that **Linear Regression** was the best-performing model, achieving an **R^2 score of approximately 84.7%**, outperforming the **Decision Tree Regressor**.

The final model is a **dependable and interpretable tool** for predicting house prices based on **location, square footage, bathrooms, and BHK**, providing valuable insights for **buyers, sellers, and real-estate professionals**.
It demonstrates practical value for **property valuation and investment analysis** in Bengaluru's housing market.

Recommendation

For similar housing datasets and price prediction tasks, **Linear Regression** is recommended due to its **high accuracy, interpretability, and consistent performance**.

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